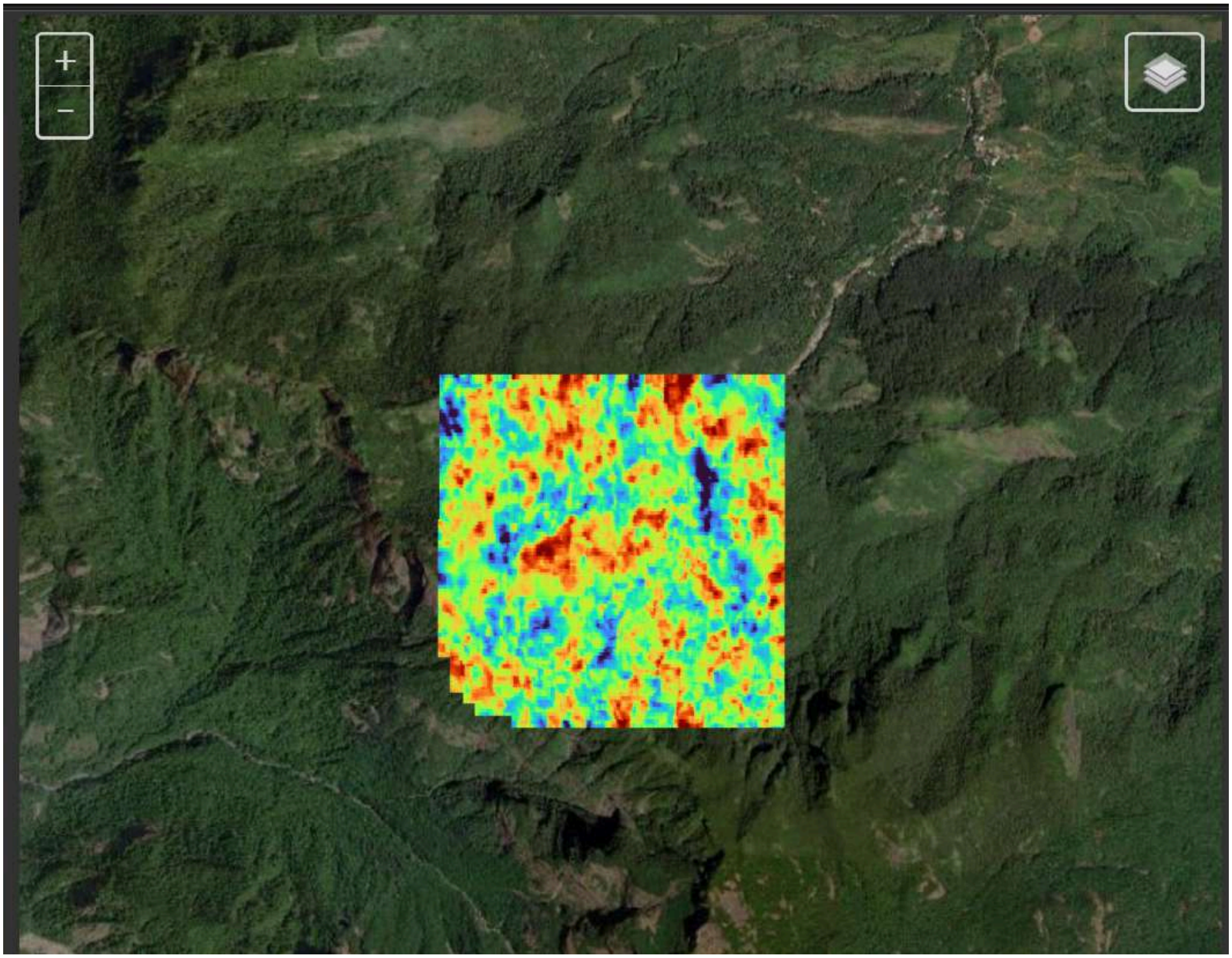
Mundakkai Hill Land subsidence Report

Mundakkai Hill (Wayanad, Kerala) Land Subsidence — InSAR Displacement Analysis Report

Abstract

This report presents a comprehensive ground deformation analysis of the Mundakkai Hill region using multi-temporal Interferometric Synthetic Aperture Radar (InSAR) measurements. Ground displacement was monitored using C-band radar acquisitions from the Sentinel-1 satellite constellation over a period spanning from January 10, 2024, to August 01, 2024. By processing the radar phase information through a stack of 16 acquisitions, we reconstructed the spatial-temporal evolution of land surface movement. The analysis reveals a highly localized subsidence pattern in the central portion of the study area, with line-of-sight velocity values ranging from approximately -30.01 mm/year to +23.67 mm/year. These results provide a detailed, millimetric spatial description of surface motion during the monitoring interval.

Area of Interest



OPTICAL SNAP SHOT - Mundakkai Hill
Bounding Box Coordinates :

[76.12683, 11.45700, 76.14517, 11.47500] # [West, South, East, North]

1. Pre-Processing & Auxiliary Data

1.1 Digital Elevation Model (DEM)

In InSAR processing, a Digital Elevation Model (DEM) is required to remove the topographic phase contribution from the interferograms. Because the radar phase is sensitive to both topography and ground motion, simulating the topographic phase contribution using a reference DEM and subtracting it from the raw interferograms is a critical step to isolate the deformation phase. For this analysis, the Copernicus Global DEM (1 arc-second, ~30m resolution) was utilized.

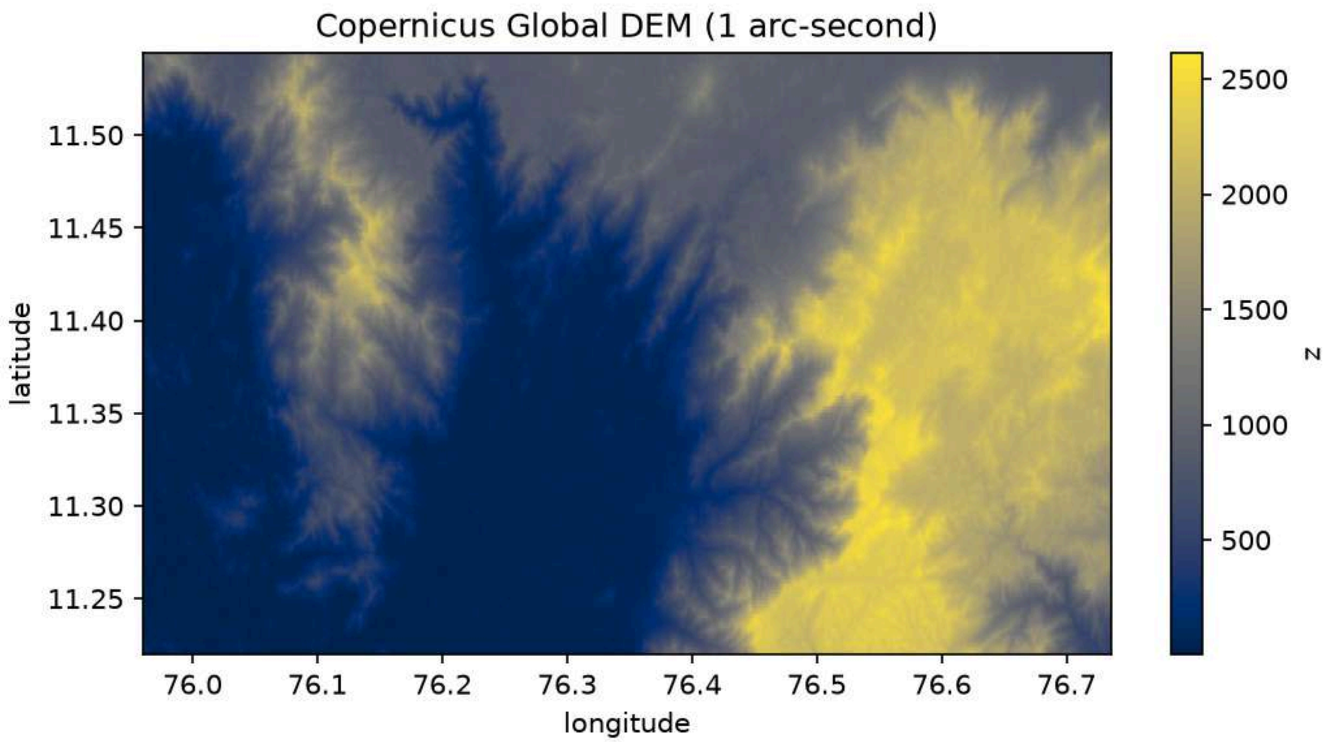


Figure 1: Copernicus Global DEM (1 arc-second) of the study area, depicting the local terrain elevation used to model and subtract topographic phase contributions.

1.2 Sentinel-1 Burst Footprints & Bounding Box

Synthetic Aperture Radar (SAR) data from Sentinel-1 is acquired in the Interferometric Wide (IW) swath mode, which consists of multiple subswaths and bursts. Individual bursts represent the basic processing units. A subset of Sentinel-1 burst footprints covering the Mundakkai Hill (Wayanad, Kerala) study area was selected to define the spatial boundaries of the analysis.

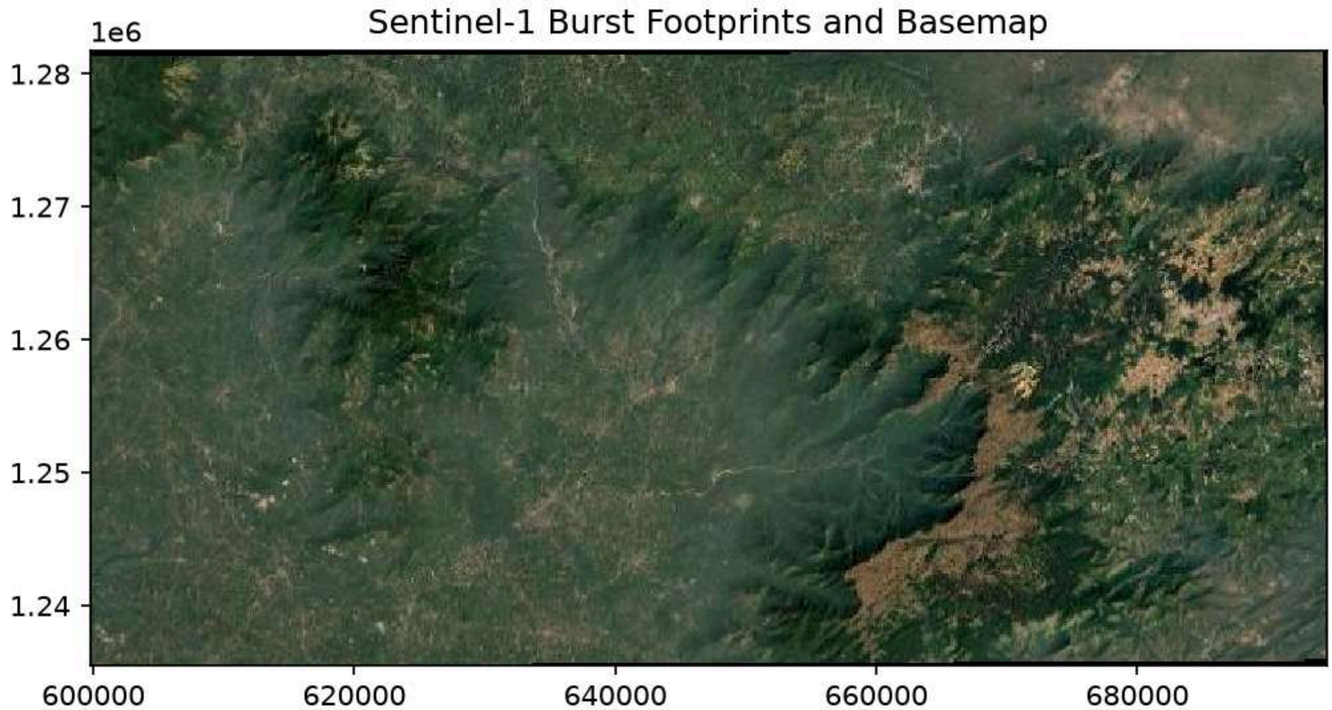


Figure 2: Sentinel-1 burst footprints overlaid on a satellite basemap, indicating the spatial coverage and footprint of the radar data relative to the study area.

1.3 Perpendicular Baseline Geometry

The perpendicular baseline ($B_\perp$) represents the spatial distance between the satellite orbit trajectories for two different acquisitions, projected perpendicular to the radar line of sight. Large perpendicular baselines introduce spatial decorrelation and topographic errors. To ensure phase coherence, a baseline threshold of 100 days (temporal) and 150 meters (spatial) was applied to select optimal interferometric pairs.

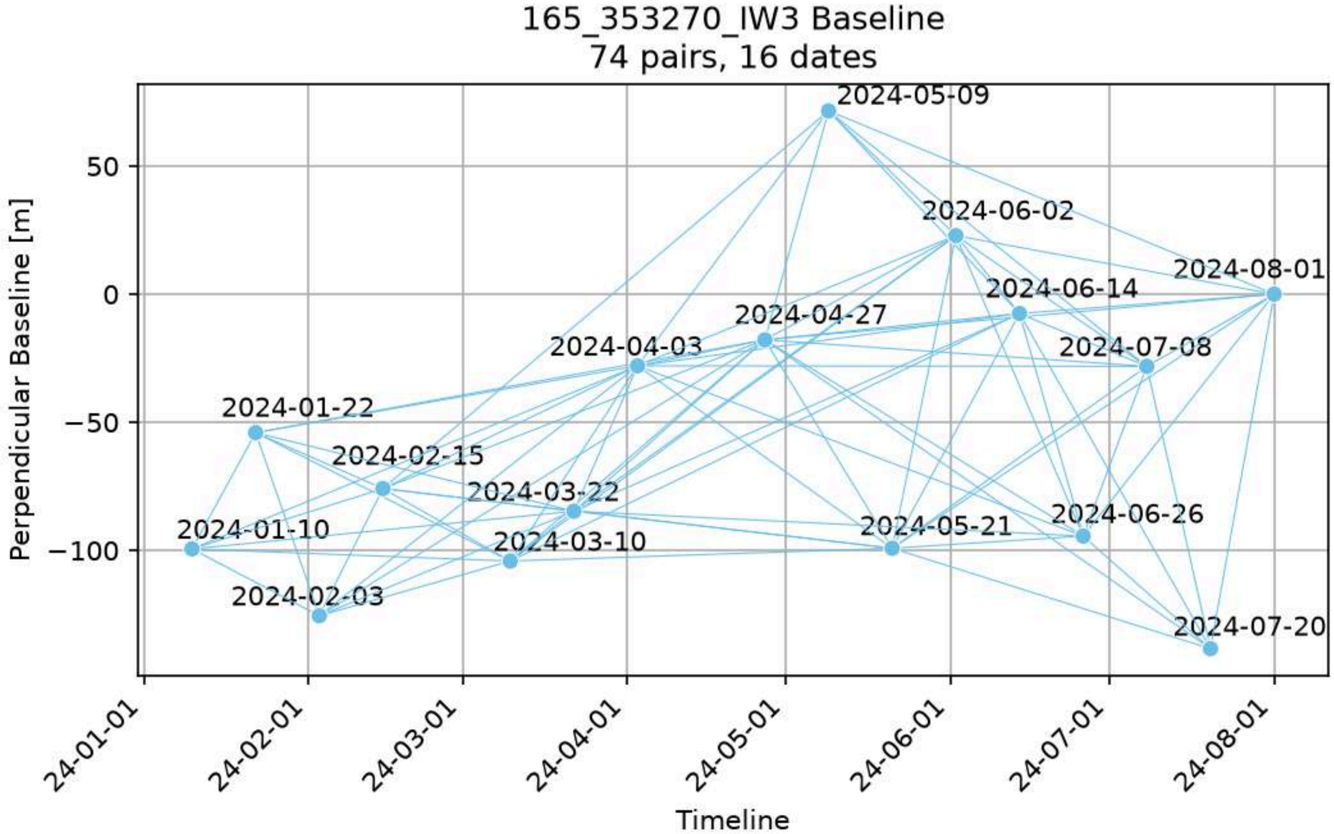


Figure 3: Perpendicular baseline distribution of the selected interferometric pairs over time relative to the reference date.

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2. Core InSAR Processing Chain

2.1 Wrapped Interferometric Phase

An interferogram is generated by cross-multiplying the complex SAR image of a reference date with the complex conjugate of a secondary date. The resulting phase represents the difference in round-trip travel distance between the two sensor positions and the ground target. This raw phase is "wrapped" within the range of $-\pi$ to $+\pi$ radians (appearing as repeated color fringes), where each full fringe cycle represents a change in sensor-to-ground distance equal to half the radar wavelength (~2.8 cm for C-band Sentinel-1).

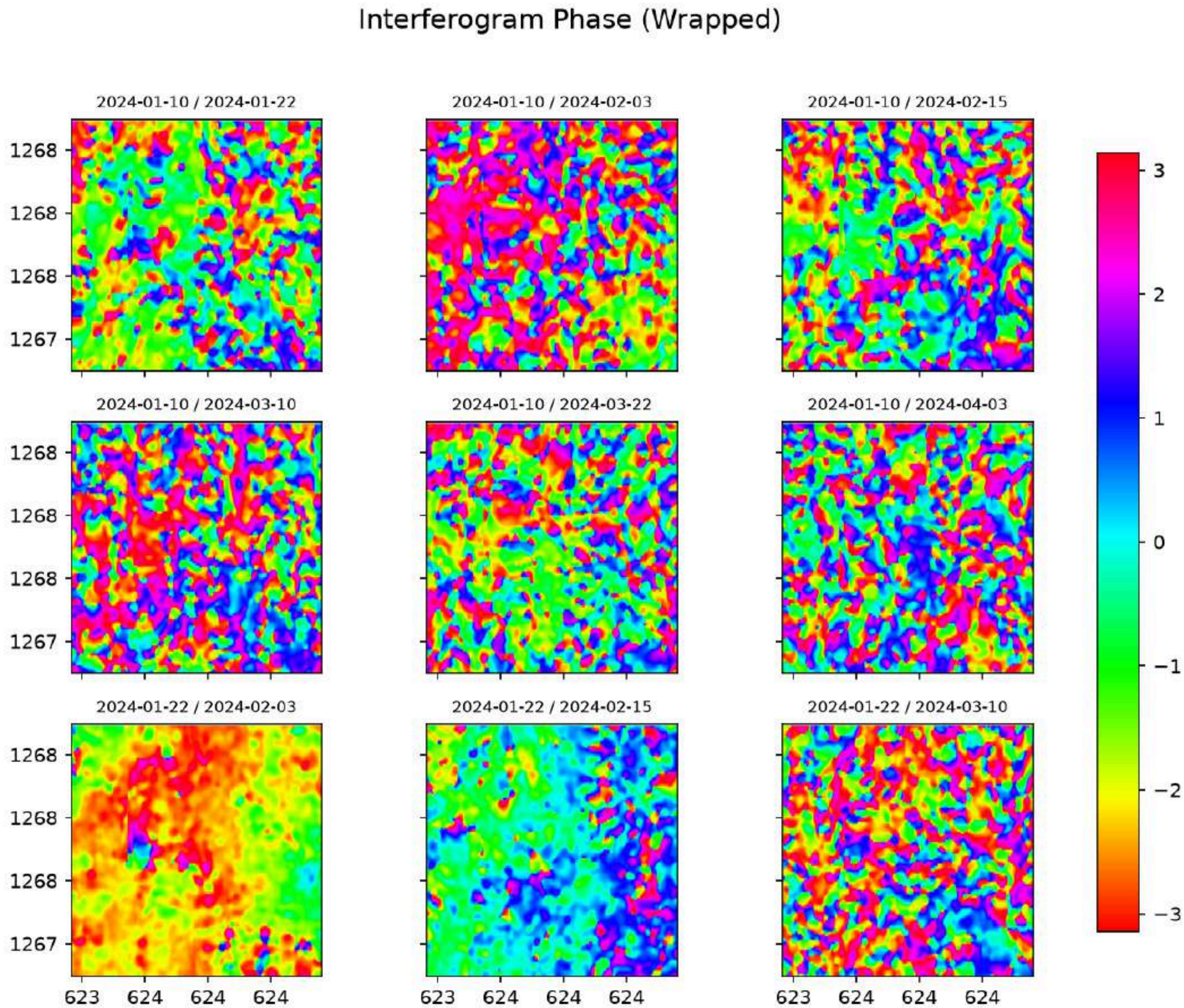

Figure 4: A 3x3 grid of wrapped interferometric phases for the first 9 selected pairs, showing characteristic wrapped fringe patterns indicating phase differences.

2.2 Interferometric Coherence (Correlation)

Coherence (or correlation) is a measure of the phase stability between two radar images. It ranges from 0 (complete noise/decorrelation) to 1 (perfect phase stability). Vegetation, open water, and human activity cause decorrelation, while dry, stable, bare surfaces maintain high coherence. Coherence maps are used to weigh and mask out noisy pixels during phase unwrapping.

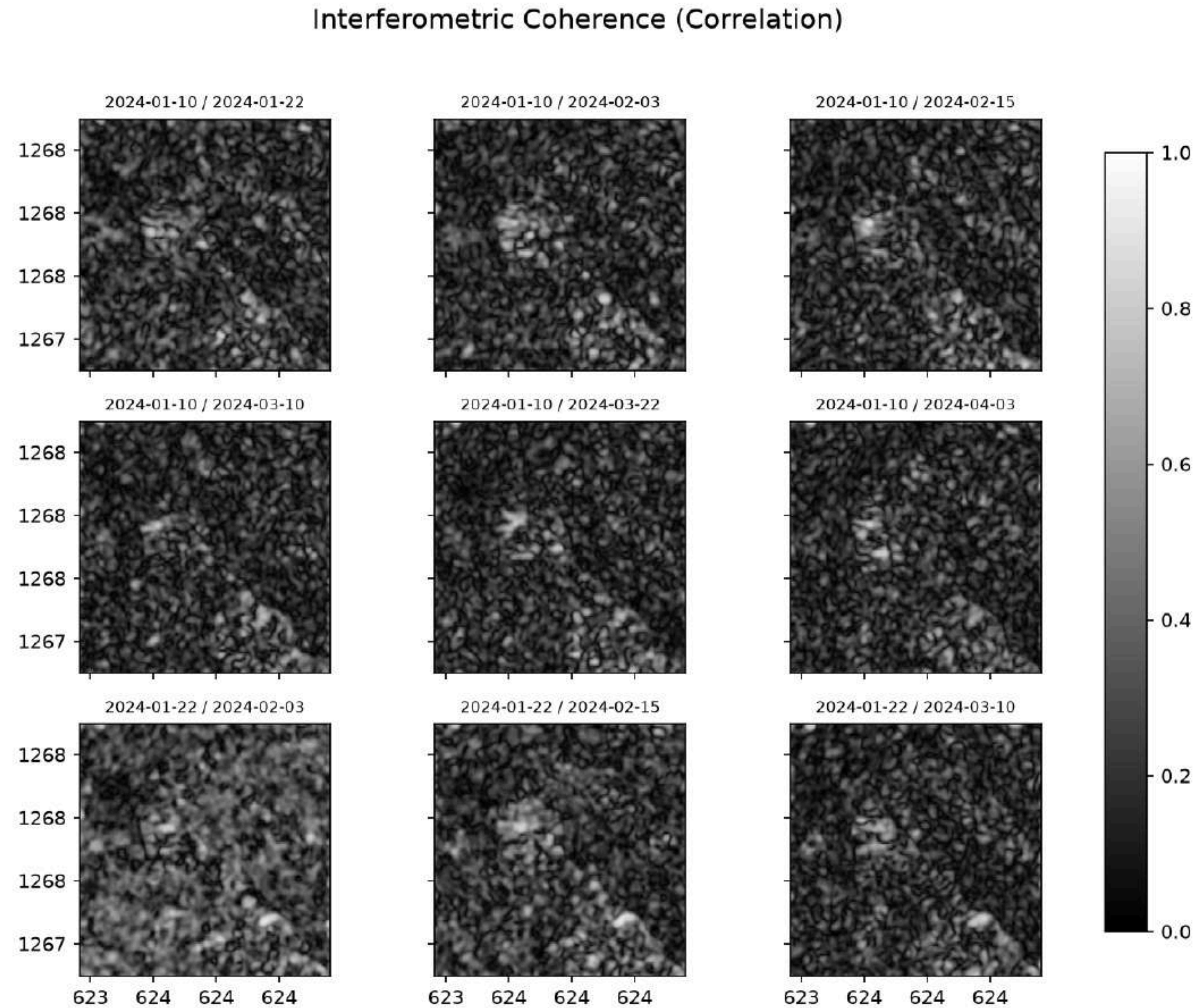

Figure 5: A 3x3 grid of coherence maps for the first 9 pairs, where bright areas represent high phase stability and dark areas represent low coherence (noise).

2.3 Unwrapped Phase

Phase unwrapping is the mathematical process of reconstructing a continuous phase field by resolving the 2π integer phase ambiguities in the wrapped interferogram. It converts the fractional phase measurements (fringes) into a continuous map of absolute phase change, which is then directly proportional to line-of-sight displacement.

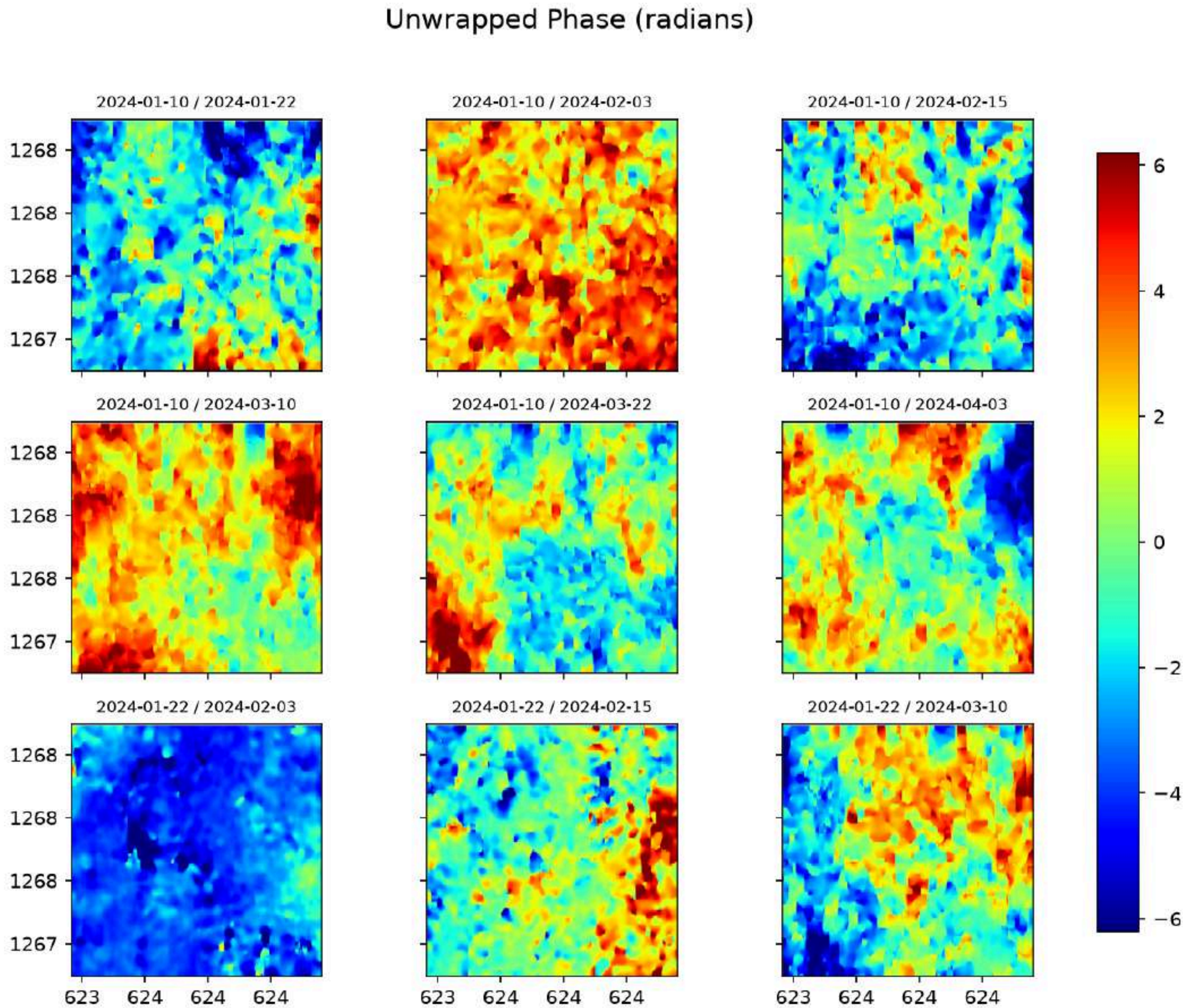

Figure 6: A 3x3 grid of unwrapped phase maps for the first 9 pairs, showing the continuous phase field after resolving 2π ambiguities.

2.4 Detrended Unwrapped Phase

Unwrapped phase maps often contain long-wavelength spatial trends (ramps) caused by minor orbital inaccuracies, ionospheric disturbances, or large-scale tropospheric delays. To isolate the localized deformation signal, a spatial detrending filter is applied. In this analysis, a Gaussian high-pass filter with a wavelength of 1.5 km was used to remove the regional trends while preserving localized ground subsidence.

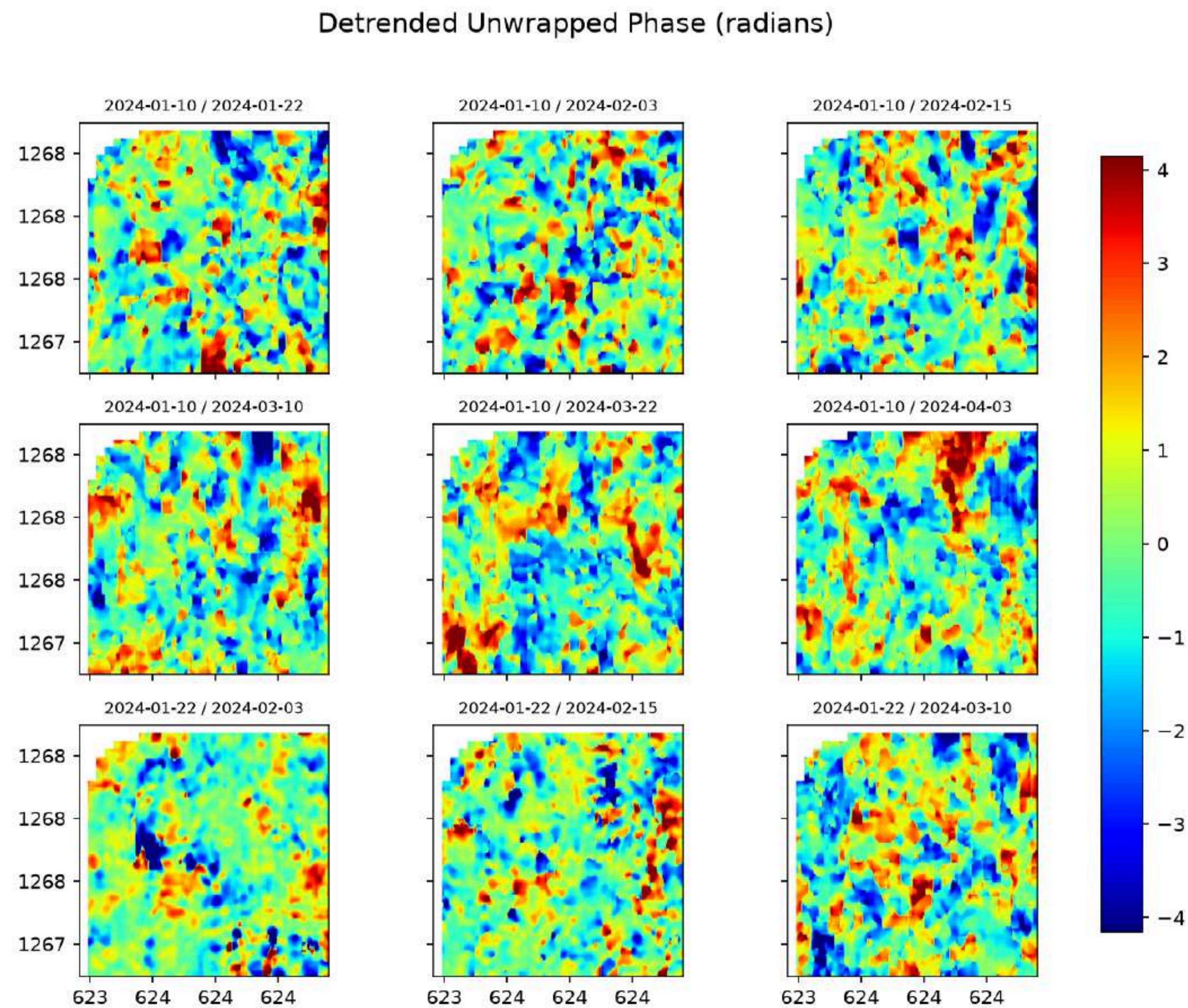

Figure 7: A 3x3 grid of spatially detrended unwrapped phases, showing the isolated localized deformation signal after removing long-wavelength noise.

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3. Mathematical Formulation

The core conversion from unwrapped phase to line-of-sight displacement is governed by:

$$\Delta\phi = \frac{4\pi}{\lambda}\, d_{LOS}$$

where $\Delta\phi$ is the unwrapped phase, λ is the radar wavelength (approximately 5.6 cm for Sentinel-1 C-band), and d_{LOS} is the displacement along the sensor's line of sight.

Cumulative displacement $d(t_i)$ is computed for each acquisition date t_i relative to the reference date t_0, where $d(t_0) = 0$.

The long-term ground velocity is estimated per pixel using a linear least-squares fit across all cumulative displacement dates:

$$d(t) \approx v \cdot t + c$$

where v is the line-of-sight velocity (slope) and c is a constant offset. The parameters are estimated by minimizing the sum of squared residuals:

$$\sum_i (d(t_i) - vt_i - c)^2$$

The native velocity units of meters per day are scaled to standard units of millimeters per year:

$$v_{mm/year} = v_{m/day} \times 365.25 \times 1000$$

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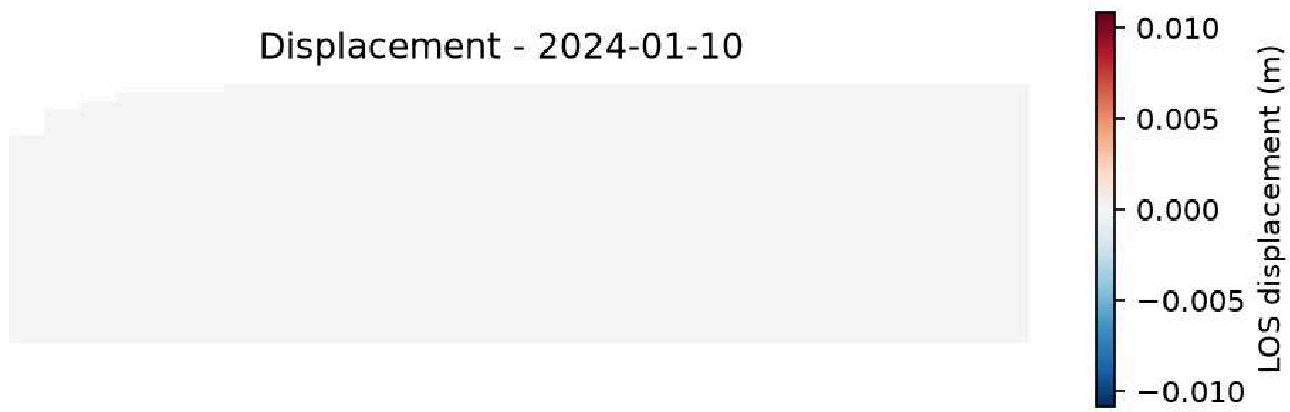
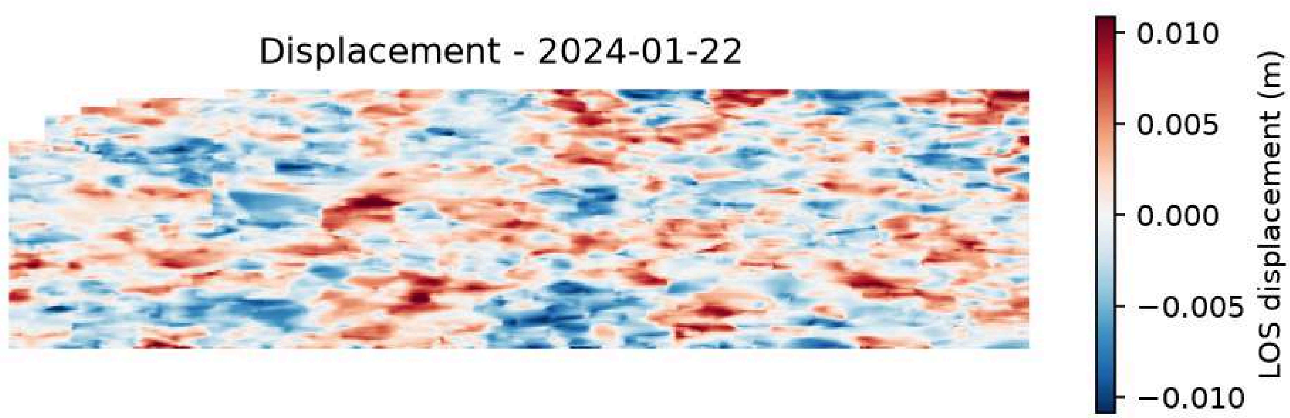
4. Cumulative Displacement Results

4.1 16-Date Cumulative Displacement Time Series

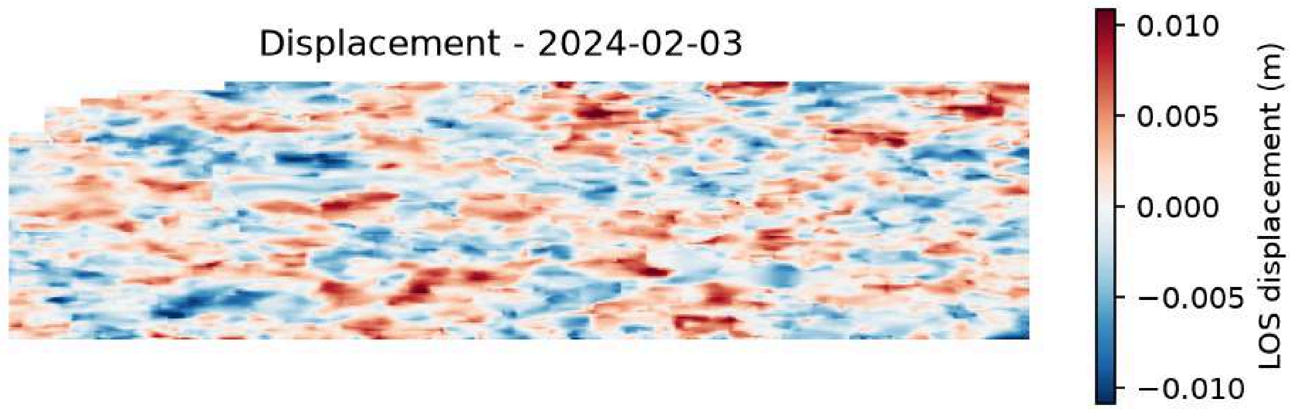
The following maps display the cumulative line-of-sight displacement (in meters) relative to the January 10, 2024 reference epoch.



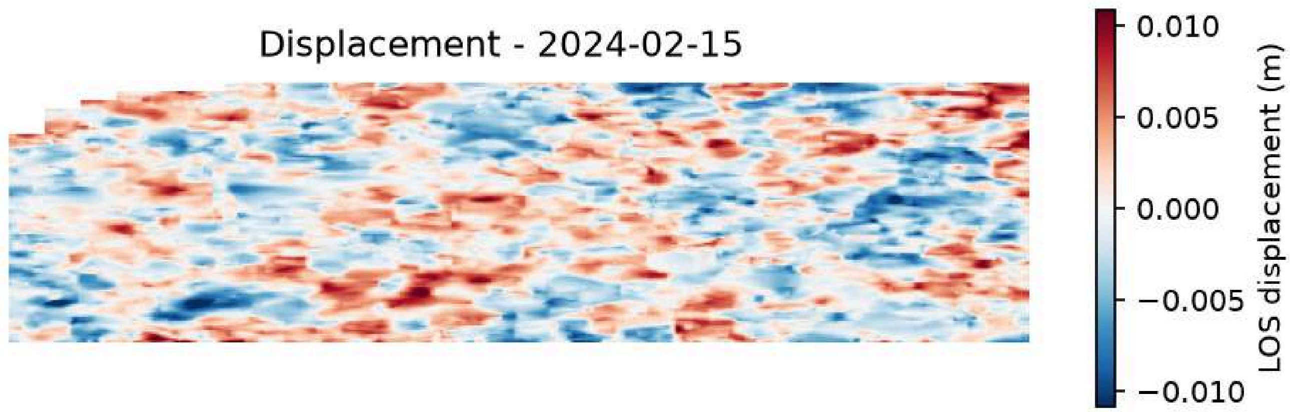
2024-01-10 (Epoch 1) — Reference date showing zero initial displacement.



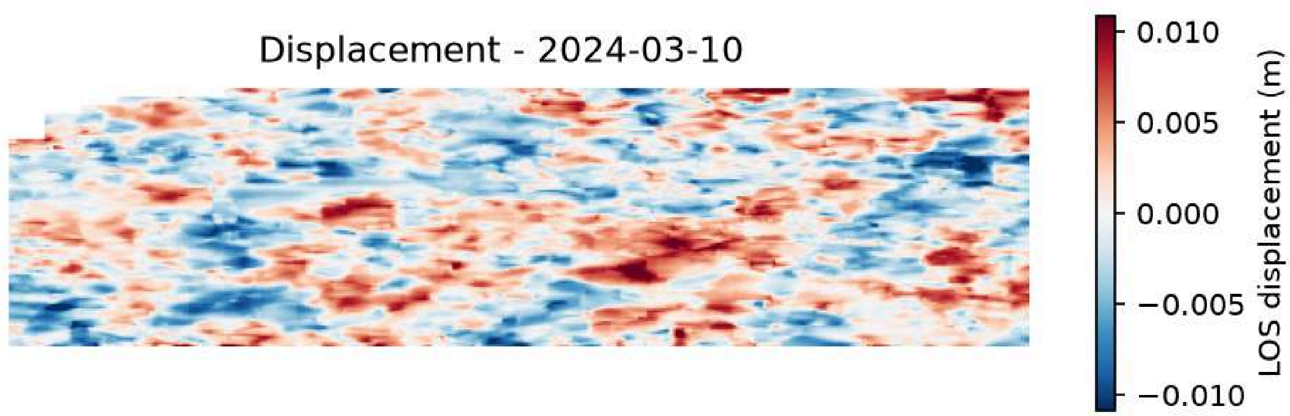
2024-01-22 (Epoch 2) — Cumulative displacement map.



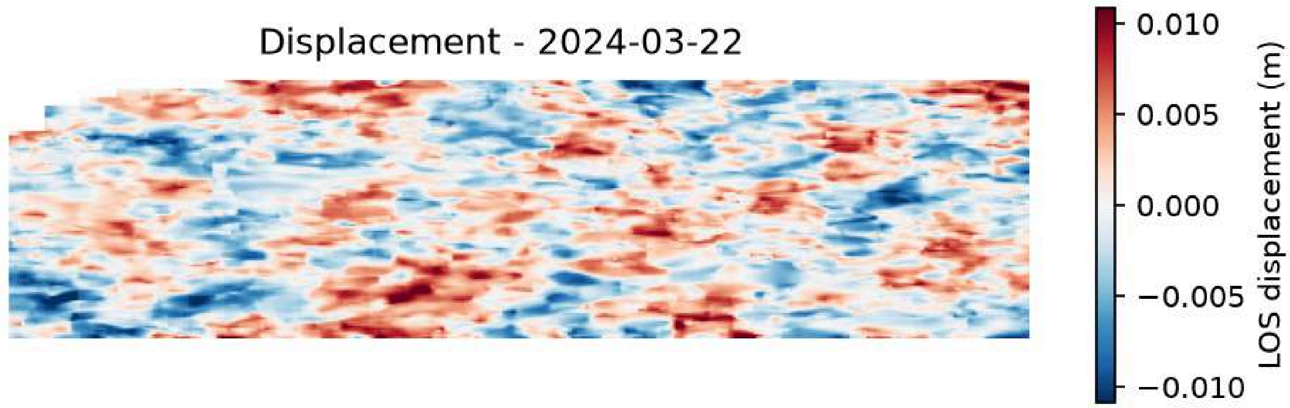
2024-02-03 (Epoch 3) — Cumulative displacement map.



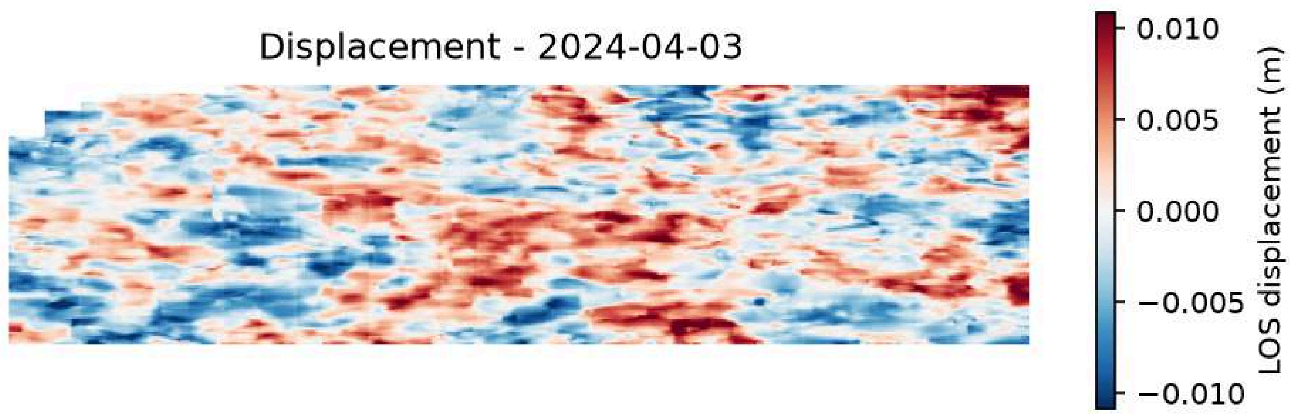
2024-02-15 (Epoch 4) — Cumulative displacement map.



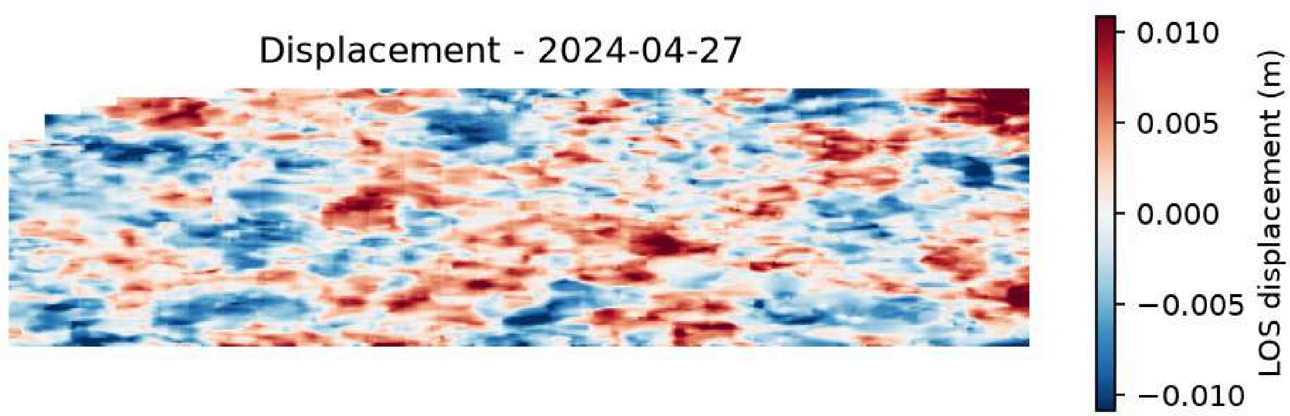
2024-03-10 (Epoch 5) — Cumulative displacement map.



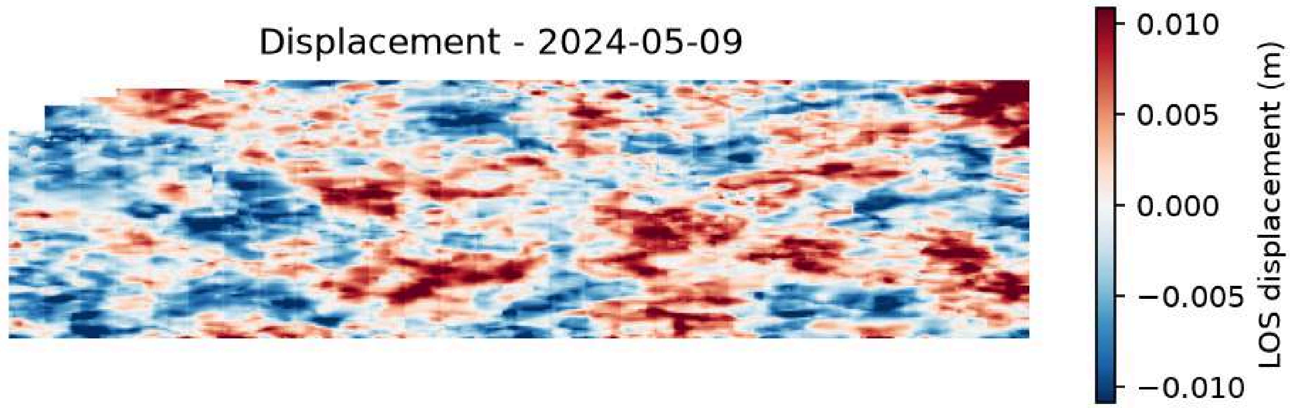
2024-03-22 (Epoch 6) — Cumulative displacement map.



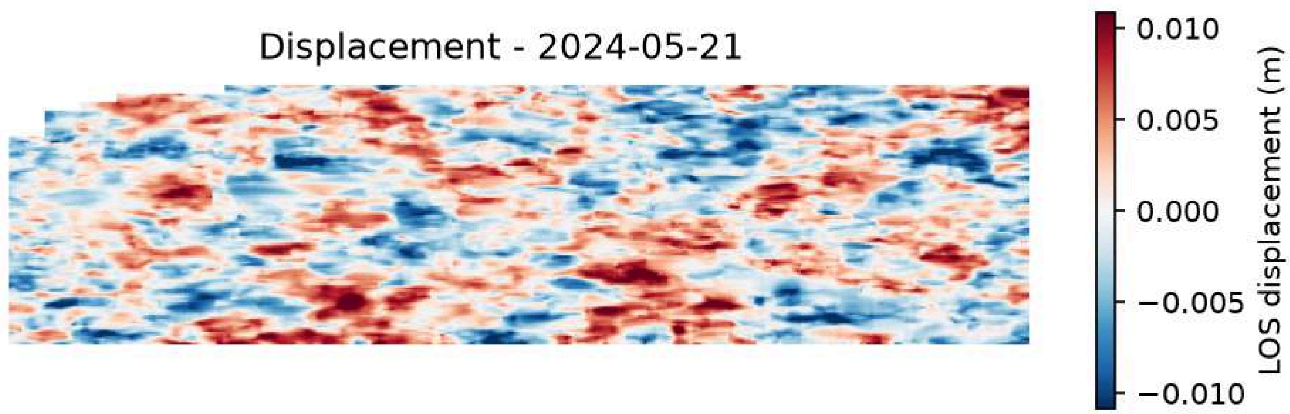
2024-04-03 (Epoch 7) — Cumulative displacement map.



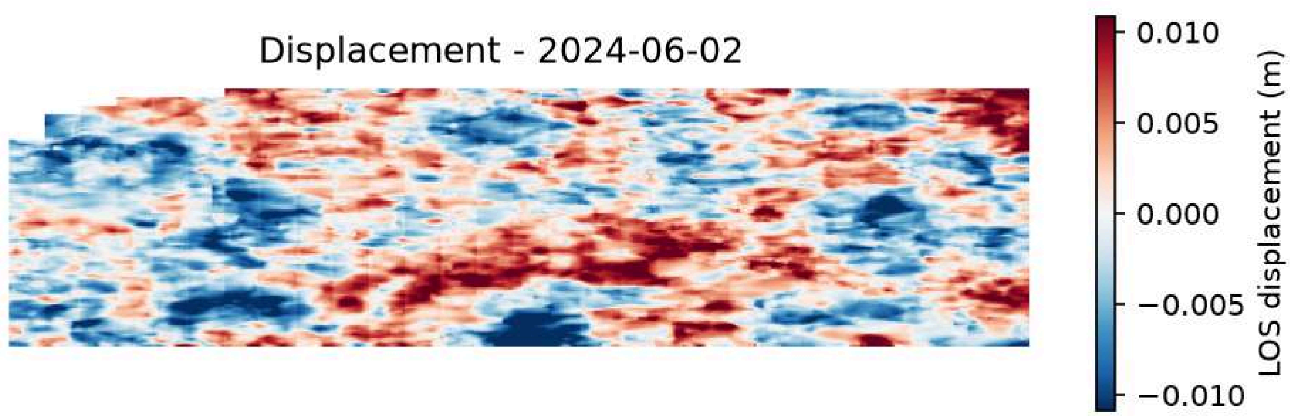
2024-04-27 (Epoch 8) — Cumulative displacement map.



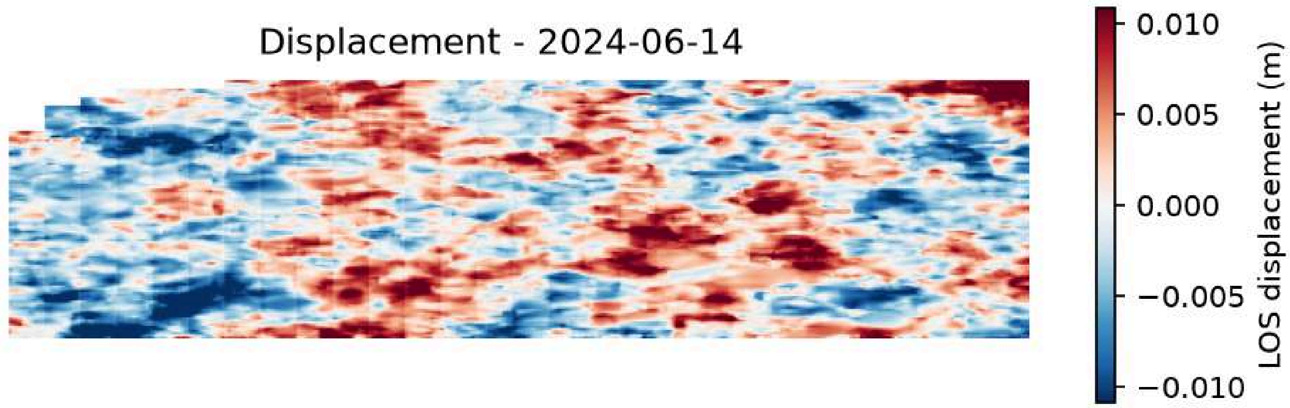
2024-05-09 (Epoch 9) — Cumulative displacement map.



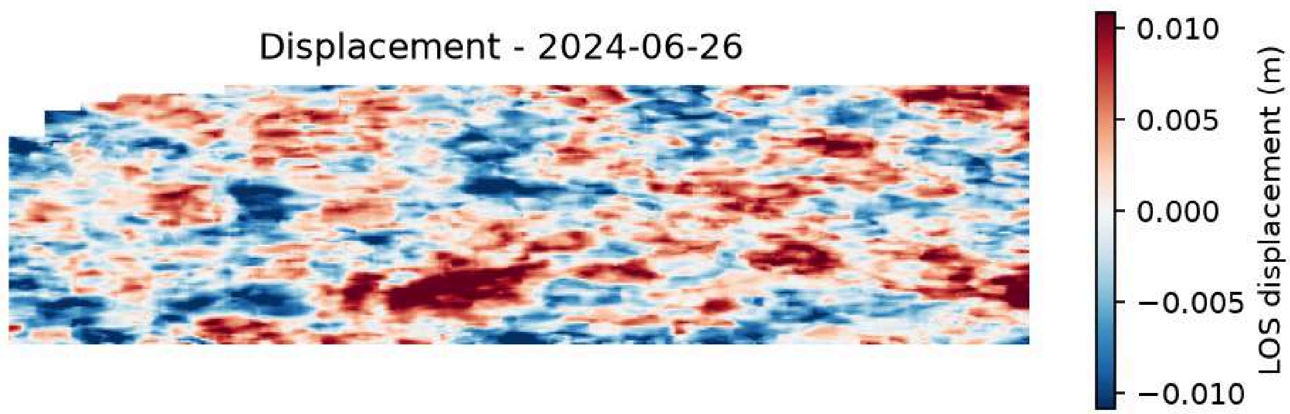
2024-05-21 (Epoch 10) — Cumulative displacement map.



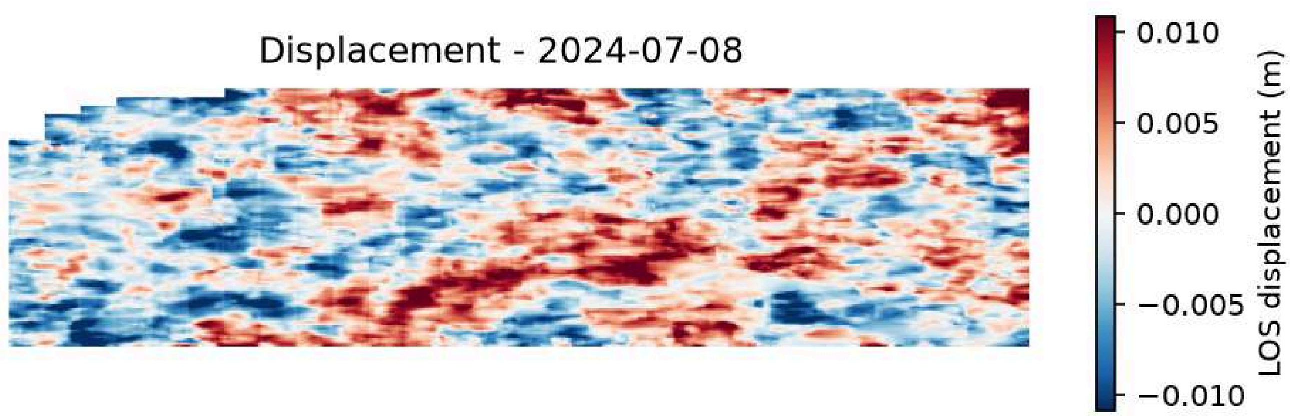
2024-06-02 (Epoch 11) — Cumulative displacement map.



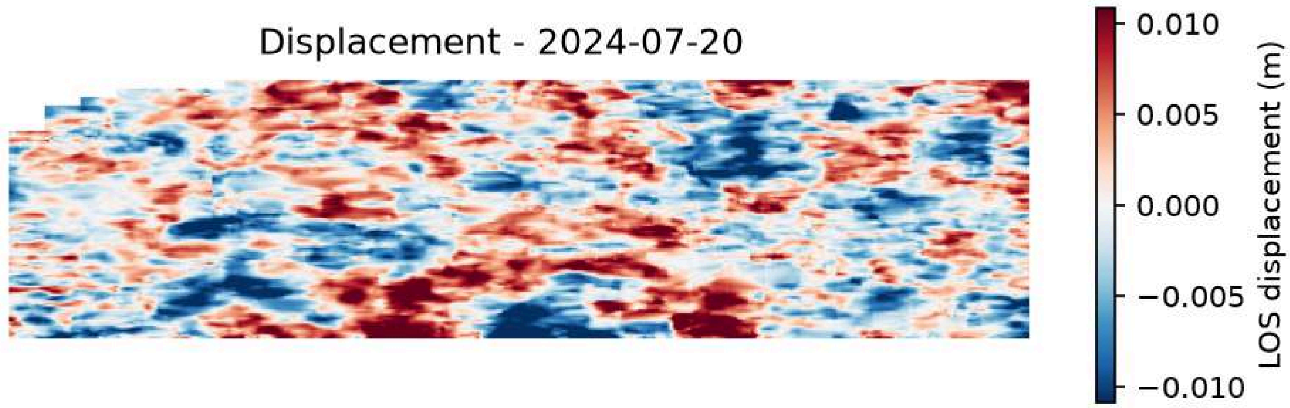
2024-06-14 (Epoch 12) — Cumulative displacement map.



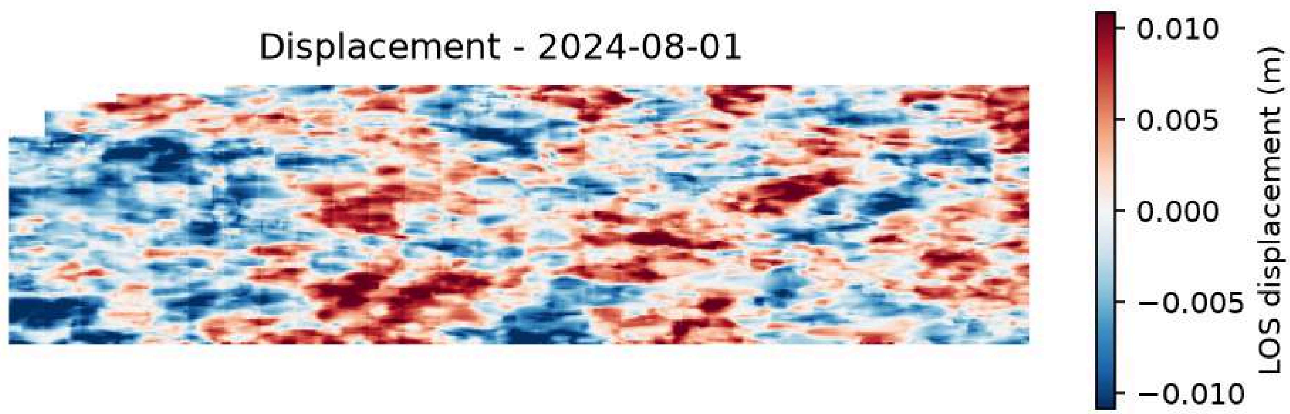
2024-06-26 (Epoch 13) — Cumulative displacement map.



2024-07-08 (Epoch 14) — Cumulative displacement map.



2024-07-20 (Epoch 15) — Cumulative displacement map.



2024-08-01 (Epoch 16) — Cumulative displacement map showing final cumulative movement over the 7-month period.

4.2 Point of Interest (POI) Cumulative Displacement

To analyze the temporal behavior of deformation, we extracted the cumulative displacement time-series at the location experiencing the maximum subsidence rate. The resulting plot shows a steady, almost linear progression of subsidence over the monitoring period.

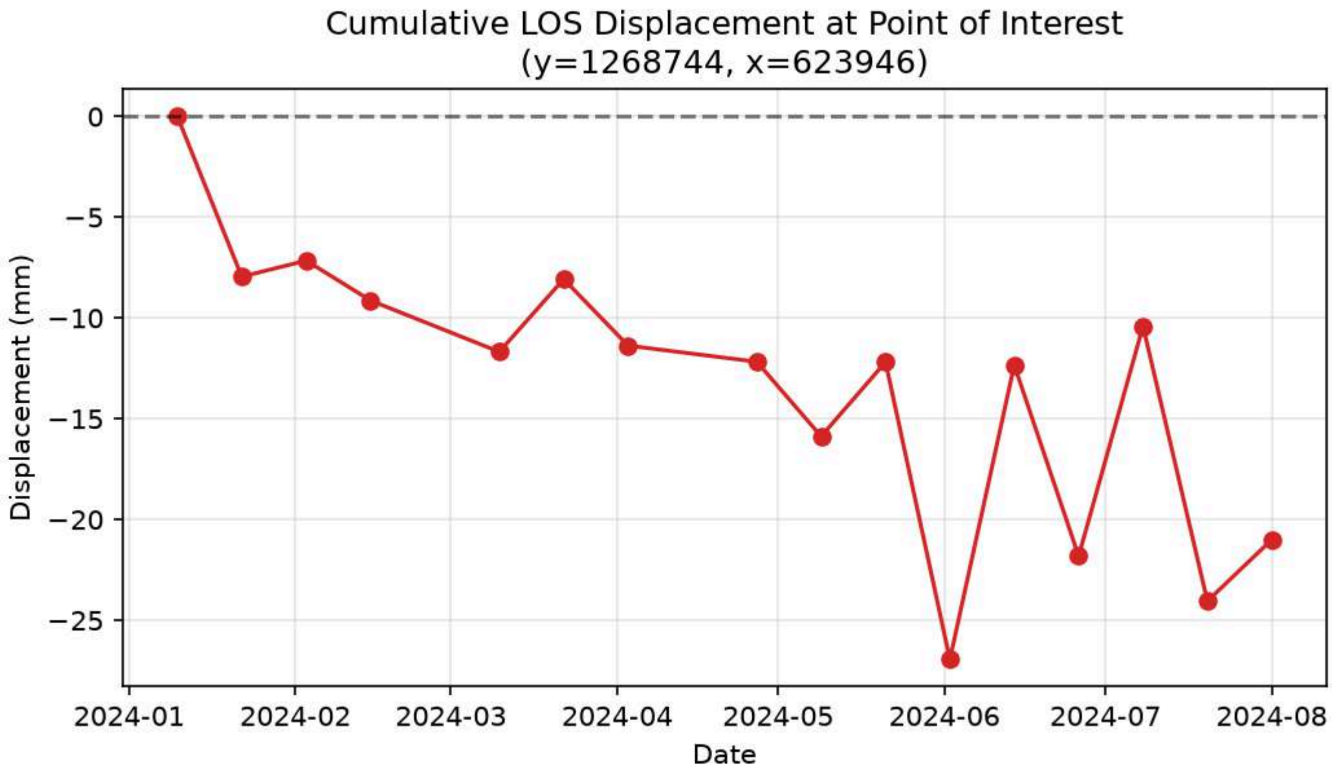


Figure 8: Time-series of cumulative line-of-sight displacement (in mm) at the point of interest (location of maximum subsidence).

4.3 Ground Velocity Map

The ground velocity map provides an estimate of the long-term deformation rate across the area. The computed line-of-sight surface velocity across the Mundakkai Hill study area ranges from a minimum of **-30.01 mm/year** to a maximum of **+23.67 mm/year**. The spatial distribution shows a clear concentric, localized pattern in the central region of the study area, surrounded by a stable background.

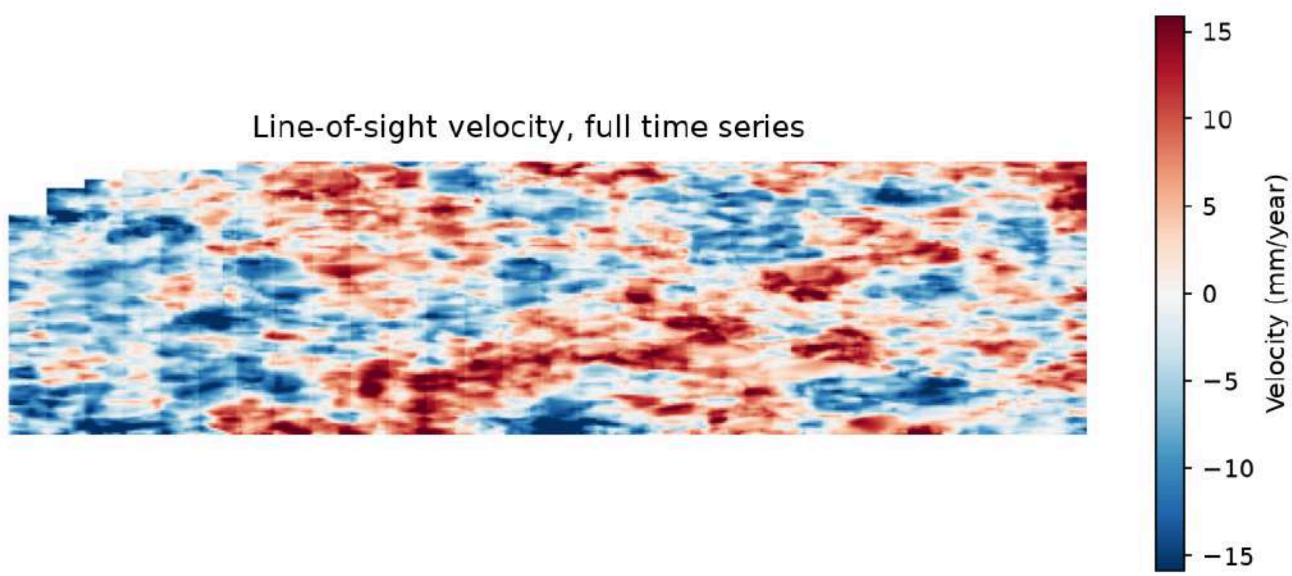


Figure 9: Line-of-sight ground velocity map (in mm/year) estimated over the full time-series from January to August 2024.

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5. Limitations

- Approximately 8% of the pixels at the borders are masked out due to coverage limits and processing boundary constraints.
- Individual single-date displacement maps are subject to noise from localized atmospheric delays and temporal decorrelation. The velocity map, which fits a trend across the entire multi-temporal series, provides a more robust estimate of long-term motion.
- The absolute direction of vertical movement (subsidence versus uplift) has not been validated against independent ground truth observations, and is reported here strictly as motion relative to the satellite's line of sight.